

Solution QUIZ 1 (Dated: 26 Sep 2024)
Data Structures and Algorithm Analysis

Roll No: _____

Name: _____

Q1. Write following codes (may write in Python) to find minimum and maximum number from list of numbers. The numbers may be negative, or both. Also, find the steps in your code.

	<u>Steps</u>
<code>min = x[0];</code>	1
<code>max = x[0];</code>	1
<code>for i in range(1, len(x)):</code>	n
<code>if min > x[i]:</code>	n - 1
<code>min = x[i]</code>	0
<code>elif max < x[i]:</code>	n - 1
<code>max = x[i]</code>	n - 1

Steps: In worst case, there will be $4n - 1$ steps.

Q2. Consider two lists of integers; write a function to return a list having elements of both lists. Make sure that elements should not occur twice in the result. Also, find the steps in your code.

	<u>Steps</u>
<code>for element2 in list2:</code>	n
<code>exist = 0</code>	n - 1
<code>for element1 in list1:</code>	$n(n - 1)$
<code>if element2 == element1:</code>	$= n^2 - n$
<code>exist = 1</code>	$(n-1)(n-1) = n^2 - 2n + 1$
<code>break</code>	
<code>if exist == 1:</code>	n - 1
<code>list1.append(element2)</code>	n - 1
<code>return list1</code>	1

Steps: In worst case, there will be $2n^2 + n - 1$ steps.

Q3. Explain the following code. Is there any better way to do the same task?

```
def f1(list1, list2):
    for i in range(len(list1)):
        for j in range(len(list2)):
            if list1[i] == list2[j]:
                print (i)
                break
```

This code is finding & printing common element in both lists. The nested loop is taking n^2 steps .

The better approach is to store list2 and for every element of list1 do binary search in list2, which will reduce the time to $n \lg n$.